WoSM Academy

M201-F27 · Calculus II

# Infinite series

## Geometric and telescoping series

Dr. Ali Khudiyev
WoSM Academy

Lecture 01 · Unit 1 · Fall 2027


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# Calculus II at a glance

**25 units in four parts**

- Units 1–8: series, power series, Taylor series
- Units 9–17: differential calculus in $\mathbb{R}^n$ and optimisation
- Units 18–22: multiple integrals
- Units 23–25: ordinary differential equations

**Assessment**

- Unit quizzes, best 10 of 12: 20%
- Project 1, series and generating functions: 15%
- Project 2, optimisation in several variables: 20%
- Final exam: 45%

Prerequisites: Calculus I (M102-S27) and Linear Algebra (M103-S27). We use limits of sequences, improper integrals and the bounds $\int f \le \sum f(k)$ from Calculus I throughout this part.

# This lecture

01 Adding infinitely many numbers

02 Series and partial sums

03 Geometric series

04 First properties

05 Telescoping series

06 The harmonic series

07 Summary and practice

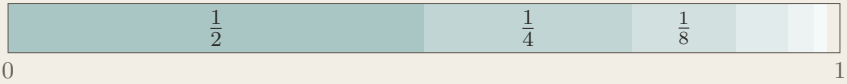
Reading: G. Strang, E. Herman, *Calculus, Volume 2* (OpenStax), Section 5.2.

01

# Adding infinitely many numbers

# Can we add infinitely many numbers?

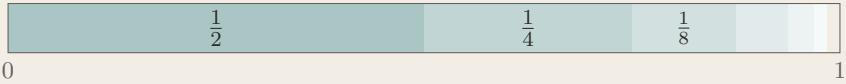
Walk halfway to a wall, then half of the remaining distance, then half again, …




After $n$ steps you have covered $\frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2^n} = 1 - \frac{1}{2^n}$, and $\frac{1}{2^n}$ is left.

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It seems right to write

$$\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \cdots = 1.$$

But the additions never finish. What does “=” mean here?

# Three answers to one question

Let $S = 1 - 1 + 1 - 1 + 1 - 1 + \cdots$. Three "proofs":

$$S = (1 - 1) + (1 - 1) + (1 - 1) + \cdots = 0 + 0 + 0 + \cdots = 0$$

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**The lesson**

The rules for finite sums — regrouping, reordering, solving for the sum — do not automatically hold for infinite sums. We need a **definition** of an infinite sum, and then **theorems** that say which manipulations are allowed.

# Where series appear in this module and later

- **Functions as series** (Units 5–6): $e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}$, $\quad \sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$
- **Counting and recurrences** (Unit 7, Project 1): generating functions
- **Approximation** (Unit 8): Stirling's formula for $n!$
- **Algorithms**: the cost of a growing array, recursion trees, the $\ln n$ in the analysis of Quicksort
- **Probability**: the probabilities of infinitely many outcomes add up to 1; expected values are series

Today: what a convergent series is, the two families of series we can sum exactly (geometric and telescoping), and the first warning sign of divergence.

02

# Series and partial sums

# Recall: sequences (Calculus I)

- $(s_n)$ **converges** to $s$ if for every $\varepsilon > 0$ there is an $N$ such that $|s_n - s| < \varepsilon$ for all $n \geq N$.
- **Limit laws**: if $s_n \to s$ and $t_n \to t$, then $s_n + t_n \to s + t$, $\; c\,s_n \to c\,s$, $\; s_n t_n \to st$.
- **Monotone convergence**: an increasing sequence that is bounded above converges (to its supremum). An increasing sequence that is not bounded above tends to $\infty$.
- **Cauchy criterion**: $(s_n)$ converges if and only if for every $\varepsilon > 0$ there is an $N$ with $|s_n - s_m| < \varepsilon$ for all $n, m \geq N$.
- If $|r| < 1$, then $r^n \to 0$.

# Series

### Definition (Series, partial sums, sum)

Let $(a_k)_{k\ge 1}$ be a sequence of real numbers. Its $n$-th **partial sum** is

$$S_n = a_1 + a_2 + \cdots + a_n = \sum_{k=1}^{n} a_k.$$

The **series** $\sum_{k=1}^{\infty} a_k$ **converges** if the sequence $(S_n)$ converges, and then its **sum** is $\displaystyle\sum_{k=1}^{\infty} a_k = \lim_{n\to\infty} S_n$. Otherwise the series **diverges**.

If $S_n \to \infty$ we write $\sum_{k=1}^{\infty} a_k = \infty$ and say the series **diverges to** $\infty$. A series *is* its sequence of partial sums, so everything we know about sequences applies.

# First examples

**Example (Halves)**

$a_k = \frac{1}{2^k}$: $S_1 = \frac{1}{2},\ S_2 = \frac{3}{4},\ S_3 = \frac{7}{8},\ \ldots,\ S_n = 1 - \frac{1}{2^n} \to 1,$ so $\sum_{k=1}^{\infty} \frac{1}{2^k} = 1.$

# First examples

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**Example (Grandi's series)**

$a_k = (-1)^{k+1}$: $\quad S_1 = 1,\ S_2 = 0,\ S_3 = 1,\ S_4 = 0,\ \ldots$ has no limit, so the series diverges.

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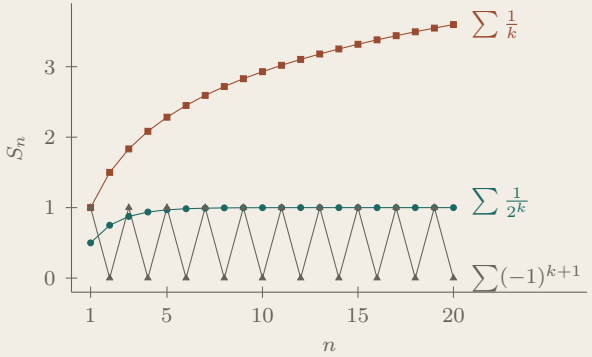
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**Example (Natural numbers)**

$a_k = k$: $S_n = \frac{n(n+1)}{2} \to \infty,$ so $\sum_{k=1}^{\infty} k = \infty.$

# Partial sums, plotted




- $\sum 1/2^k$: the partial sums settle at 1.
- $\sum(-1)^{k+1}$: they jump between 1 and 0 forever.
- $\sum 1/k$: they keep growing, ever more slowly. Do they settle? (Section 6.)

# Notation and the starting index

- A series may start at any index: $\displaystyle\sum_{k=0}^{\infty} r^k$, $\displaystyle\sum_{k=2}^{\infty} \frac{1}{k^2-1}$. Partial sums then start there too.
- Shifting the index changes nothing: $\displaystyle\sum_{k=1}^{\infty} a_k = \sum_{j=0}^{\infty} a_{j+1}$ (put $j = k-1$).
- The name of the index does not matter: $\sum_k a_k = \sum_j a_j$.
- When the range is clear we write $\sum a_k$.

**Warning**

The symbol $\sum_{k=1}^{\infty} a_k$ stands for two things: the series (the sequence of partial sums) and, if it converges, its sum (a number). Writing $\sum_{k=1}^{\infty} a_k = S$ *claims* that the series converges.

03

# Geometric series

# Finite geometric sums

**Lemma**

For $r \neq 1$ and $n \geq 0$:

$$\sum_{k=0}^{n} r^k = 1 + r + r^2 + \cdots + r^n = \frac{1 - r^{n+1}}{1 - r}.$$

# Finite geometric sums

**Lemma**

For $r \neq 1$ and $n \geq 0$:

$$\sum_{k=0}^{n} r^k = 1 + r + r^2 + \cdots + r^n = \frac{1 - r^{n+1}}{1 - r}.$$

**Proof.**

Let $G_n = 1 + r + \cdots + r^n$. Then $rG_n = r + r^2 + \cdots + r^{n+1}$, so in $G_n - rG_n = 1 - r^{n+1}$ every other term cancels — our first *telescoping* sum (Section 5). Divide by $1 - r \neq 0$. □

# The geometric series

**Theorem (Geometric series)**

$\sum_{k=0}^{\infty} r^k$ converges if and only if $|r| < 1$, and then $\sum_{k=0}^{\infty} r^k = \frac{1}{1-r}$.

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**Proof.**

Use $G_n = \frac{1-r^{n+1}}{1-r}$ for $r \neq 1$.

- $|r| < 1$: $r^{n+1} \to 0$, so $G_n \to \frac{1}{1-r}$.
- $r = 1$: $G_n = n + 1 \to \infty$. $r = -1$: $G_n = 1, 0, 1, 0, \ldots$ has no limit.
- $|r| > 1$: $|G_n| \geq \frac{|r|^{n+1}-1}{|1-r|} \to \infty$: unbounded, so divergent. □

# First term over one minus the ratio

**Corollary**

If $|r| < 1$, then for any number $a$ and any integer $m$

$$\sum_{k=m}^{\infty} ar^k = \frac{ar^m}{1-r} = \frac{\text{first term}}{1-\text{ratio}}.$$

Proof: the partial sums are $ar^m G_n$ (put $j = k - m$); use the limit laws.

To use it: identify the first term and the ratio $r$, and check that $|r| < 1$.

# Examples

$$\sum_{k=1}^{\infty} \frac{3}{4^k}$$

$$\sum_{k=0}^{\infty} \left(-\frac{2}{3}\right)^k$$

$$\sum_{k=2}^{\infty} \frac{2^{k+1}}{5^k}$$

$$\sum_{k=0}^{\infty} \left(\frac{3}{2}\right)^k$$

# Examples

$$\sum_{k=1}^{\infty} \frac{3}{4^k} = \frac{3/4}{1 - 1/4} = 1$$

$$\sum_{k=0}^{\infty} \left(-\frac{2}{3}\right)^k = \frac{1}{1 + 2/3} = \frac{3}{5}$$

$$\sum_{k=2}^{\infty} \frac{2^{k+1}}{5^k} = \frac{8/25}{1 - 2/5} = \frac{8}{15}$$

$$\sum_{k=0}^{\infty} \left(\frac{3}{2}\right)^k \text{ diverges: } |r| = \tfrac{3}{2} \geq 1$$

# Examples

$$\sum_{k=1}^{\infty} \frac{3}{4^k} = \frac{3/4}{1 - 1/4} = 1 \qquad \sum_{k=0}^{\infty} \left(-\frac{2}{3}\right)^k = \frac{1}{1 + 2/3} = \frac{3}{5}$$

$$\sum_{k=2}^{\infty} \frac{2^{k+1}}{5^k} = \frac{8/25}{1 - 2/5} = \frac{8}{15} \qquad \sum_{k=0}^{\infty} \left(\frac{3}{2}\right)^k \text{ diverges: } |r| = \tfrac{3}{2} \geq 1$$

**Common mistakes**

The first term is not always the coefficient: in $\sum_{k=2}^{\infty} \frac{2^{k+1}}{5^k}$ it is $\frac{2^3}{5^2} = \frac{8}{25}$. And $\frac{1}{1-r}$ is the sum only when the series starts at $r^0 = 1$.

# Repeating decimals

**Example**

$$0.999\ldots = \frac{9}{10} + \frac{9}{100} + \frac{9}{1000} + \cdots = \frac{9/10}{1 - 1/10} = 1.$$

- The same argument shows that every repeating decimal is a rational number.
- A decimal expansion *is* a series: $0.d_1d_2d_3\ldots = \sum_{k=1}^{\infty} d_k\, 10^{-k}$.
- In binary, $0.0111\ldots_2 = \frac{1}{4} + \frac{1}{8} + \cdots = \frac{1}{2} = 0.1_2$: some numbers have two expansions.

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# Application: an array that doubles

When a growable array is full, a common implementation allocates one of twice the capacity and copies every element across: capacities $1, 2, 4, 8, \ldots$

- Inserting $n$ elements copies $1, 2, 4, \ldots, 2^j$ elements, where $2^j$ is the largest power of 2 below $n$. In total, reading from the largest term:

$$2^j\Big(1 + \frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2^j}\Big) < 2^j \sum_{i=0}^{\infty} \frac{1}{2^i} = 2 \cdot 2^j < 2n.$$

  So $n$ insertions cost $O(n)$ in total: $O(1)$ per insertion, amortised.

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  So $n$ insertions cost $O(n)$ in total: $O(1)$ per insertion, amortised.

> **Remark**
>
> Terms that shrink geometrically sum to at most a constant times the largest one. So $T(n) = T(n/2) + n$ gives $T(n) \le n\,(1 + \frac{1}{2} + \frac{1}{4} + \cdots) = 2n$.

# Application: waiting for a success

Repeat independent trials, each succeeding with probability $p$, $0 < p < 1$.

$$P(\text{first success at trial } k) = (1-p)^{k-1}\,p, \qquad k = 1, 2, 3, \ldots$$

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These probabilities form a geometric series with ratio $1 - p$, and $|1 - p| < 1$:

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So, with probability 1, a success eventually happens.

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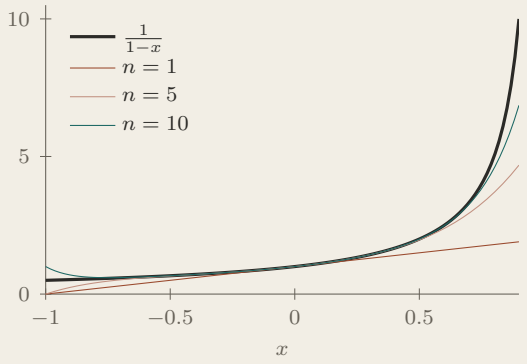
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> **Later in the module**
>
> The expected number of trials is $\sum_{k=1}^{\infty} k\,(1-p)^{k-1}p = \frac{1}{p}$. Summing it needs $\sum_{k\geq 1} kx^{k-1} = \frac{1}{(1-x)^2}$, proved in Lecture 4.

# A first power series




For $-1 < x < 1$:

$$\frac{1}{1-x} = \sum_{k=0}^{\infty} x^k = 1 + x + x^2 + \cdots$$

and the series diverges for $|x| \geq 1$.

The partial sums $1 + x + \cdots + x^n$ approach $\frac{1}{1-x}$ for every $x$ in $(-1, 1)$ — more slowly the closer $x$ is to $\pm 1$. A function written as a series in powers of $x$ is a *power series*: Unit 5.

04

# First properties

# Linearity

**Theorem**

If $\sum a_k = A$ and $\sum b_k = B$ both converge and $c \in \mathbb{R}$, then

$$\sum (a_k + b_k) = A + B \qquad \text{and} \qquad \sum c\, a_k = cA.$$

Proof: the partial sums are $A_n + B_n$ and $cA_n$; use the limit laws.

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Proof: the partial sums are $A_n + B_n$ and $cA_n$; use the limit laws.

**Example**

$$\sum_{k=0}^{\infty} \frac{2^k + 3^k}{6^k} = \sum_{k=0}^{\infty} \left(\frac{1}{3}\right)^k + \sum_{k=0}^{\infty} \left(\frac{1}{2}\right)^k = \frac{3}{2} + 2 = \frac{7}{2}.$$

# Splitting a series

> **Warning**
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> Linearity works one way only: $\sum(1 + (-1)) = 0$ converges, but $\sum 1$ and $\sum(-1)$ diverge. Split a series only when you know both parts converge.

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**Corollary**

If $\sum a_k$ converges and $\sum b_k$ diverges, then $\sum(a_k+b_k)$ diverges.

**Proof.**

If $\sum(a_k+b_k)$ converged, then by linearity so would $\sum((a_k+b_k)-a_k)=\sum b_k$. □

For example, $\sum\left(\frac{1}{2^k}+1\right)$ diverges.

# Only the tail matters

**Theorem**

Let $m \geq 1$. Then $\sum_{k=1}^{\infty} a_k$ converges if and only if $\sum_{k=m+1}^{\infty} a_k$ converges, and then

$$\sum_{k=1}^{\infty} a_k = \sum_{k=1}^{m} a_k + \sum_{k=m+1}^{\infty} a_k.$$

Proof: for $n > m$, $S_n = S_m + \sum_{k=m+1}^{n} a_k$, and $S_m$ is a fixed number.

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Proof: for $n > m$, $S_n = S_m + \sum_{k=m+1}^{n} a_k$, and $S_m$ is a fixed number.

- Changing, adding or removing *finitely many* terms does not change whether a series converges. It may change the sum.
- If $\sum a_k = S$, the **remainder** $R_n = \sum_{k=n+1}^{\infty} a_k = S - S_n$ tends to 0. The error of stopping after $n$ terms is $R_n$.

# The $n$-th term test

**Theorem**

If $\sum a_k$ converges, then $a_k \to 0$.

**Proof.**

$a_n = S_n - S_{n-1} \to S - S = 0.$ □

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**Corollary (Divergence test)**

If $(a_k)$ does not converge to 0, then $\sum a_k$ diverges.

# Using the $n$-th term test

**Example**

These all diverge, because their terms do not tend to 0:

- $\sum \frac{k}{k+1}$ (terms $\to 1$) $\qquad$ $\sum k \sin \frac{1}{k}$ (terms $\to 1$)
- $\sum (-1)^k$ (terms have no limit) $\qquad$ $\sum r^k$ with $|r| \geq 1$ ($|r^k| \geq 1$)

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**Warning**

The converse is false: $a_k \to 0$ does *not* imply that $\sum a_k$ converges (Sections 5 and 6). The test can only prove divergence.

# Grouping terms

**Theorem**

If $\sum a_k$ converges to $S$, then inserting parentheses — grouping consecutive terms into blocks — gives a series that also converges to $S$.

Proof: the partial sums of the grouped series are $S_{n_1}, S_{n_2}, S_{n_3}, \dots$ for some $n_1 < n_2 < \cdots$: a subsequence of $(S_n)$, which has the same limit.

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So for $1-1+1-\cdots$: the first two "proofs" grouped a *divergent* series, and the third assumed that the sum exists. None of 0, 1, $\frac{1}{2}$ is its sum.

# The Cauchy criterion

**Theorem (Cauchy criterion for series)**

$\sum a_k$ converges if and only if for every $\varepsilon > 0$ there is an $N$ such that

$$|a_{m+1} + a_{m+2} + \cdots + a_n| < \varepsilon \qquad \text{for all } n > m \geq N.$$

**Proof.**

$a_{m+1} + \cdots + a_n = S_n - S_m$, and $(S_n)$ converges if and only if it is a Cauchy sequence (Calculus I, Unit 4). □

- It proves convergence *without knowing the sum.* Lecture 3 uses it for absolutely convergent series.
- With $n = m + 1$ it gives the $n$-th term test again.

05

# Telescoping series

# Telescoping sums

If each term is a difference of consecutive terms of another sequence, $a_k = b_k - b_{k+1}$, the partial sums collapse:

$$S_n = (b_1 - b_2) + (b_2 - b_3) + \cdots + (b_n - b_{n+1}) = b_1 - b_{n+1}.$$

# Telescoping sums

If each term is a difference of consecutive terms of another sequence, $a_k = b_k - b_{k+1}$, the partial sums collapse:

$$S_n = (b_1 - b_2) + (b_2 - b_3) + \cdots + (b_n - b_{n+1}) = b_1 - b_{n+1}.$$

**Theorem (Telescoping series)**

$\sum_{k=1}^{\infty}(b_k - b_{k+1})$ converges if and only if $(b_n)$ converges, and then

$$\sum_{k=1}^{\infty}(b_k - b_{k+1}) = b_1 - \lim_{n\to\infty} b_n.$$

Proof: $S_n = b_1 - b_{n+1}$. The finite geometric sum used the same cancellation.

# Example: $\sum \frac{1}{k(k+1)}$

Partial fractions (Calculus I): $\displaystyle \frac{1}{k(k+1)} = \frac{1}{k} - \frac{1}{k+1}$, so $b_k = \frac{1}{k}$.

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and in general $S_n = 1 - \frac{1}{n+1} \to 1$.

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$$\sum_{k=1}^{\infty} \frac{1}{k(k+1)} = \frac{1}{2} + \frac{1}{6} + \frac{1}{12} + \frac{1}{20} + \cdots = 1.$$

We found the sum because $S_n$ has a closed form. Usually it does not, and we can only decide *whether* a series converges (Lecture 2).

# Example: a gap of two

$$\sum_{k=2}^{\infty} \frac{1}{k^2-1}, \qquad \frac{1}{k^2-1} = \frac{1}{(k-1)(k+1)} = \frac{1}{2}\Big(\frac{1}{k-1} - \frac{1}{k+1}\Big).$$

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In $\sum_{k=2}^{n}\big(\frac{1}{k-1} - \frac{1}{k+1}\big)$ the positive terms are $1, \frac{1}{2}, \frac{1}{3}, \dots, \frac{1}{n-1}$ and the negative terms are $\frac{1}{3}, \dots, \frac{1}{n-1}, \frac{1}{n}, \frac{1}{n+1}$. Two survive at each end:

$$S_n = \frac{1}{2}\Big(1 + \frac{1}{2} - \frac{1}{n} - \frac{1}{n+1}\Big) \longrightarrow \frac{1}{2}\cdot\frac{3}{2} = \frac{3}{4}.$$

# Example: three factors

$$\frac{1}{k(k+1)(k+2)} = \frac{1}{2}\Big(\frac{1}{k(k+1)} - \frac{1}{(k+1)(k+2)}\Big)$$

(check: the bracket is $\frac{(k+2)-k}{k(k+1)(k+2)}$).

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(check: the bracket is $\frac{(k+2)-k}{k(k+1)(k+2)}$). So the term is $b_k - b_{k+1}$ with $b_k = \frac{1}{2k(k+1)}$, and $b_n \to 0$:

$$\sum_{k=1}^{\infty} \frac{1}{k(k+1)(k+2)} = b_1 - \lim_{n\to\infty} b_n = \frac{1}{4}.$$

# Example: telescoping series that diverge

**Example**

$$\sum_{k=1}^{\infty} \ln\Big(1+\frac{1}{k}\Big) = \sum_{k=1}^{\infty} \big(\ln(k+1) - \ln k\big): \qquad S_n = \ln(n+1) - \ln 1 = \ln(n+1) \to \infty.$$

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$$\sum_{k=1}^{\infty} (\sqrt{k+1} - \sqrt{k}): \qquad S_n = \sqrt{n+1} - 1 \to \infty.$$

In both the terms tend to 0: $\ln(1+\frac{1}{k}) \to 0$ and $\sqrt{k+1} - \sqrt{k} = \frac{1}{\sqrt{k+1}+\sqrt{k}} \to 0$.
They are our first divergent series with $a_k \to 0$.

# Always work with partial sums

## Two wrong arguments

- $\sum\left(\frac{1}{k}-\frac{1}{k+1}\right)=\sum\frac{1}{k}-\sum\frac{1}{k+1}$: both series on the right diverge (Section 6), and $\infty-\infty$ means nothing. Linearity needs both parts to converge.
- "In $\sum\big(\ln(k+1)-\ln k\big)$ every term cancels, so the sum is $-\ln 1=0$." In fact $S_n=\ln(n+1)\to\infty$.

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- "In $\sum\big(\ln(k+1) - \ln k\big)$ every term cancels, so the sum is $-\ln 1 = 0$." In fact $S_n = \ln(n+1) \to \infty$.

**Recipe**

1. Write $a_k = b_k - b_{k+1}$ (partial fractions, rules for logarithms, rationalising, …).
2. Compute $S_n$ exactly: which terms survive?
3. Take the limit of $S_n$.

06

# The harmonic series

# The harmonic series diverges

The $n$-th **harmonic number** is $H_n = \sum_{k=1}^{n} \frac{1}{k}$.

> **Theorem**
>
> $$\sum_{k=1}^{\infty} \frac{1}{k} = \infty.$$

The terms tend to 0, so the $n$-th term test says nothing. The partial sums grow (Section 2's plot), but do they stay below some bound? Two proofs that they do not.

# Oresme's proof (14th century)

**Proof.**

Group the terms in blocks that end at powers of 2:

$$1 + \tfrac{1}{2} + \underbrace{\left(\tfrac{1}{3} + \tfrac{1}{4}\right)}_{\geq 2\cdot\frac{1}{4}} + \underbrace{\left(\tfrac{1}{5} + \cdots + \tfrac{1}{8}\right)}_{\geq 4\cdot\frac{1}{8}} + \underbrace{\left(\tfrac{1}{9} + \cdots + \tfrac{1}{16}\right)}_{\geq 8\cdot\frac{1}{16}} + \cdots$$

The block that ends at $\frac{1}{2^j}$ has $2^{j-1}$ terms, each at least $\frac{1}{2^j}$, so it adds at least $\frac{1}{2}$. Hence $H_{2^m} \geq 1 + \frac{m}{2} \to \infty$. As $(H_n)$ is increasing, $H_n \to \infty$. □

# A second proof

For $x > -1$, $\ln(1+x) \le x$ (the graph of $\ln(1+x)$ lies below its tangent at 0).

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$$H_n = \sum_{k=1}^{n} \frac{1}{k} \;\ge\; \sum_{k=1}^{n} \ln\!\Big(1 + \frac{1}{k}\Big) = \ln(n+1) \;\to\; \infty,$$

using the telescoping sum from Section 5.

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using the telescoping sum from Section 5.

> **Remark**
>
> We compared $\sum \frac{1}{k}$ with a series that we already knew diverges, term by term. Lecture 2 turns this idea into a test: the *comparison test*.

# How fast does $H_n$ grow?

From Calculus I (Unit 26):

$$\ln(n+1) \le H_n \le 1 + \ln n.$$

| $n$ | $\ln(n+1)$ | $H_n$ | $1+\ln n$ |
|---|---|---|---|
| 10 | 2.398 | 2.929 | 3.303 |
| 100 | 4.615 | 5.187 | 5.605 |
| 1000 | 6.909 | 7.485 | 7.908 |
| $10^6$ | 13.816 | 14.393 | 14.816 |

- $H_n$ grows like $\ln n$: without bound, but very slowly.
- The first $n$ with $H_n > 10$ is $n = 12\,367$.
- In fact $H_n - \ln n \to \gamma \approx 0.5772$, the Euler–Mascheroni constant.

# Harmonic numbers in algorithms

- **Quicksort** with a uniformly random pivot, on $n$ distinct keys, makes on average

$$2(n+1)H_n - 4n \;\approx\; 2n\ln n \quad \text{comparisons.}$$

- **Coupon collector**: draw uniformly at random, with replacement, from $n$ coupons. The expected number of draws until every coupon has been seen is

$$nH_n \;\approx\; n\ln n.$$

In both, the logarithm comes from a harmonic sum $\sum_{k=1}^{n} \frac{1}{k}$.

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$$S_n = 1 + \sum_{k=2}^{n} \frac{1}{k^2} < 1 + \sum_{k=2}^{n} \Big( \frac{1}{k-1} - \frac{1}{k} \Big) = 2 - \frac{1}{n} < 2.$$

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An increasing sequence that is bounded above converges, so $\sum \frac{1}{k^2}$ converges, and its sum is at most 2.

> **Remark**
>
> Euler found the sum: $\sum_{k=1}^{\infty} \frac{1}{k^2} = \frac{\pi^2}{6} \approx 1.6449$ (the Basel problem). Our argument proves convergence without finding the sum.

# Looking ahead

In both $\sum \frac{1}{k}$ and $\sum \frac{1}{k^2}$ the terms tend to 0, yet one diverges and the other converges. Where is the boundary?

$$\sum \frac{1}{k^{1.01}}\,? \qquad \sum \frac{1}{\sqrt{k}}\,? \qquad \sum \frac{1}{k \ln k}\,? \qquad \sum \frac{k^{10}}{2^k}\,? \qquad \sum \frac{k!}{k^k}\,?$$

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**The key fact for Lecture 2**

If $a_k \geq 0$ for all $k$, the partial sums increase, so

$$\sum a_k \text{ converges} \iff (S_n) \text{ is bounded above.}$$

Lecture 2 turns this into practical tests: comparison, integral, ratio and root.

07

# Summary and practice

# Summary

- $\sum a_k$ **converges** if its partial sums $S_n$ converge; the sum is $\lim S_n$.
- **Geometric**: $\sum_{k=0}^{\infty} r^k = \frac{1}{1-r}$ if $|r| < 1$, and it diverges if $|r| \geq 1$. In general: first term $/(1 - \text{ratio})$.
- **Linearity** holds for convergent series. Finitely many terms never decide convergence.
- $n$**-th term test**: if $\sum a_k$ converges, then $a_k \to 0$. It can only prove divergence.
- **Grouping** a convergent series is allowed; removing parentheses is not.
- **Telescoping**: $\sum(b_k - b_{k+1}) = b_1 - \lim b_n$, if the limit exists.
- **Harmonic series**: $\sum \frac{1}{k} = \infty$, with $H_n \approx \ln n$; but $\sum \frac{1}{k^2}$ converges (bounded partial sums).

# Practice

1. Find $\displaystyle\sum_{k=1}^{\infty}(-1)^{k+1}\frac{5}{3^k}$.
2. Write $0.142857142857\ldots$ as a fraction.
3. Find $\displaystyle\sum_{k=1}^{\infty}\frac{1}{4k^2-1}$.
4. Find $\displaystyle\sum_{k=1}^{\infty}\ln\frac{k(k+2)}{(k+1)^2}$.

# Practice, continued

5. Do these converge? (a) $\displaystyle\sum \frac{k^2}{2k^2+1}$ (b) $\displaystyle\sum \Big(1+\frac{1}{k}\Big)^{k}$
6. For which $x$ does $\sum_{k=0}^{\infty}(x-1)^k$ converge? Find its sum.
7. True or false: if $\sum a_k$ and $\sum b_k$ both diverge, then $\sum (a_k + b_k)$ diverges.
8. Show that if $\sum a_k$ converges, then $\sum_{k=n+1}^{\infty} a_k \to 0$ as $n \to \infty$.
9. A ball is dropped from a height of $2\,\text{m}$. After each bounce it rises to $\frac{3}{5}$ of its previous height. How far does it travel in total?

# Answers

1. $\frac{5}{4}$ (first term $\frac{5}{3}$, ratio $-\frac{1}{3}$).
2. $\frac{142857}{999999} = \frac{1}{7}$.
3. $\frac{1}{2}$, since $\frac{1}{4k^2-1} = \frac{1}{2}\left(\frac{1}{2k-1} - \frac{1}{2k+1}\right)$.
4. $-\ln 2$, with $b_k = \ln \frac{k}{k+1}$: the term is $b_k - b_{k+1}$, $b_1 = -\ln 2$, $b_n \to 0$.
5. Both diverge: the terms tend to $\frac{1}{2}$ and to $e$.
6. For $0 < x < 2$ (ratio $x - 1$); the sum is $\frac{1}{2-x}$.
7. False: $a_k = 1$, $b_k = -1$.
8. The tail is $S - S_n$, and $S_n \to S$.
9. $$2 + 2 \cdot \frac{2 \cdot \frac{3}{5}}{1 - \frac{3}{5}} = 2 + 6 = 8\,\text{m}.$$

# Next time

**Lecture 02 — Convergence tests** (Unit 2)

- Series with nonnegative terms
- The integral test and $p$-series
- Comparison and limit comparison
- The ratio and root tests

Reading: Strang & Herman, *Calculus, Volume 2*, Sections 5.3, 5.4 and 5.6.

Questions: the module channel on Discord, or office hours, Monday–Friday, 10:00–17:00 (Baku time, UTC+4), ali.khudiyev@wosm.academy.